

A Practical Guide to Text summarization

TOPIC -- TEXT SUMMARIZATION

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Abstractive-based summarization Abstractive summarization methods generate new text that did not exist in the original text. This has been applied mainly for text. Abstractive methods build an internal semantic representation of the original content (often called a language model), and then use this representation to create a summary that is closer to what a human might express. Abstraction may transform the extracted content by paraphrasing sections of the source document, to condense a text more strongly than extraction. Such transformation, however, is computationally much more challenging than extraction, involving both natural language processing and often a deep understanding of the domain of the original text in cases where the original document relates to a special field of knowledge. "Paraphrasing" is even more difficult to apply to images and videos, which is why most summarization systems are extractive.

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TextRank and LexRank The unsupervised approach to summarization is also quite similar in spirit to unsupervised keyphrase extraction and gets around the issue of costly training data. Some unsupervised summarization approaches are based on finding a "centroid" sentence, which is the mean word vector of all the sentences in the document. Then the sentences can be ranked with regard to their similarity to this centroid sentence. A more principled way to estimate sentence importance is using random walks and eigenvector centrality. LexRank is an algorithm essentially identical to TextRank, and both use this approach for document summarization. The two methods were developed by different groups at the same time, and LexRank simply focused on summarization, but could just as easily be used for keyphrase extraction or any other NLP ranking task. In both LexRank and TextRank, a graph is constructed by creating a vertex for each sentence in the document. The edges between sentences are based on some form of semantic similarity or content overlap. While LexRank uses cosine similarity of TF-IDF vectors, TextRank uses a very similar measure based on the number of words two sentences have in common (normalized by the sentences' lengths). The LexRank paper explored using unweighted edges after applying a threshold to the cosine values, but also experimented with using edges with weights equal to the similarity score. TextRank uses continuous similarity scores

as weights. In both algorithms, the sentences are ranked by applying PageRank to the resulting graph. A summary is formed by combining the top ranking sentences, using a threshold or length cutoff to limit the size of the summary. It is worth noting that TextRank was applied to summarization exactly as described here, while LexRank was used as part of a larger summarization system (MEAD) that combines the LexRank score (stationary probability) with other features like sentence position and length using a linear combination with either user-specified or automatically tuned weights. In this case, some training documents might be needed, though the TextRank results show the additional features are not absolutely necessary. Unlike TextRank, LexRank has been applied to multi-document summarization.

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The Reddit bot "autotldr", created in 2011 summarizes news articles in the comment-section of reddit posts. It was found to be very useful by the reddit community which upvoted its summaries hundreds of thousands of times. The name is reference to TL;DR – Internet slang for "too long; didn't read". Adversarial stylometry may make use of summaries, if the detail lost is not major and the summary is sufficiently stylistically different to the input. Automatic summarization has been extended to audio content such as podcasts, where systems combine speech recognition, large language model summarization, and speech synthesis to produce condensed narrated audio summaries of full-length episodes.

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"The Army Corps of Engineers, rushing to meet President Bush's promise to protect New Orleans by the start of the 2006 hurricane season, installed defective flood-control pumps last year despite warnings from its own expert that the equipment would fail during a storm, according to documents obtained by The Associated Press". A keyphrase extractor might select "Army Corps of Engineers", "President Bush", "New Orleans", and "defective flood-control pumps" as keyphrases. These are pulled directly from the text. In contrast, an abstractive keyphrase system would somehow internalize the content and generate keyphrases that do not appear in the text, but more closely resemble what a human might produce, such as "political negligence" or "inadequate protection from floods". Abstraction requires a deep understanding of the text, which makes it difficult for a computer system. Keyphrases have many applications. They can enable document browsing by providing a short summary, improve information retrieval (if documents have keyphrases assigned, a user could search by keyphrase to produce more reliable hits than a

full-text search), and be employed in generating index entries for a large text corpus. Depending on the different literature and the definition of key terms, words or phrases, keyword extraction is a highly related theme.

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Applications and systems for summarization There are broadly two types of extractive summarization tasks depending on what the summarization program focuses on. The first is generic summarization, which focuses on obtaining a generic summary or abstract of the collection (whether documents, or sets of images, or videos, news stories etc.). The second is query relevant summarization, sometimes called query-based summarization, which summarizes objects specific to a query. Summarization systems are able to create both query relevant text summaries and generic machine-generated summaries depending on what the user needs. An example of a summarization problem is document summarization, which attempts to automatically produce an abstract from a given document. Sometimes one might be interested in generating a summary from a single source document, while others can use multiple source documents (for example, a cluster of articles on the same topic). This problem is called multi-document summarization. A related application is summarizing news articles. Imagine a system, which automatically pulls together news articles on a given topic (from the web), and concisely represents the latest news as a summary. Image collection summarization is another application example of automatic summarization. It consists in selecting a representative set of images from a larger set of images. A summary in this context is useful to show the most representative images of results in an image collection exploration system. Video summarization is a related domain, where the system automatically creates a trailer of a long video. This also has applications in consumer or personal videos, where one might want to skip the boring or repetitive actions. Similarly, in surveillance videos, one would want to extract important and suspicious activity, while ignoring all the boring and redundant frames captured. At a very high level, summarization algorithms try to find subsets of objects (like set of sentences, or a set of images), which cover information of the entire set. This is also called the core-set. These algorithms model notions like diversity, coverage, information and representativeness of the summary. Query based summarization techniques, additionally model for relevance of the summary with the query. Some techniques and algorithms which naturally model summarization problems are TextRank and PageRank, Submodular set function, Determinantal point process, maximal marginal relevance (MMR) etc.